When History Can Replace Action Chunking

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Abstract

We study when history replaces action chunking in stable linear imitation learning. Observability permits exact noiseless imitation, but restricted data need not identify it. We combine a history-based chunk-stability bound, a conditional certificate for general learners, and an explicit ridge stability certificate. Recovery experiments support shorter execution with longer history without establishing universal monotonicity.

1. History supplies information; learning selects feedback

Consider $x_{t+1} = Ax_t + Bu_t$, $y_t = Cx_t$, and expert $u_t^* = Kx_t$, with $F = A + BK$. Restricted demonstrations can conceal recovery behavior even for stable plants and experts (Simchowitz et al., 2025); commitment can suppress destabilizing learned feedback (Zhang et al., 2025). We ask whether history reduces the need for that commitment.

The learner observes $z_t^{(r)} = \mathrm{col}(y_{t-r+1:t}, u_{t-r+1:t-1})$. Context r counts observations, prediction horizon H counts predicted actions, and execution $\ell \le H$ counts actions committed before replanning. Let $h = \min\{r : \operatorname{rank} \mathcal{O}_r = n\}$, where $\mathcal{O}_r = \mathrm{col}(C, CA, \ldots, CA^{r-1})$. Reconstruction assumes noiseless measurements, known applied inputs, known dynamics, and a valid initial history.

Proposition 1 (action recovery and execution equivalence). An exact current-action policy exists on all noiseless histories iff

$$\ker \mathcal{O}_r \subseteq \ker(KA^{r-1}).$$

For $r \ge h$, state reconstruction is $x_t = R_r z_t$, with

$$R_r = [A^{r-1}\mathcal{O}_r^\dagger \quad \mathcal{R}_r - A^{r-1}\mathcal{O}_r^\dagger \mathcal{T}_r],$$

where $Y = \mathcal{O}_r x_{t-r+1} + \mathcal{T}_r U$ and $x_t = A^{r-1}x_{t-r+1} + \mathcal{R}_r U$ define the usual input maps. The chunk policy $\pi_k^*(z) = KF^k R_r z$ then reproduces the expert for every execution length $1 \le \ell \le H$.

Proof. Identical histories with identical inputs differ initially by $d \in \ker \mathcal{O}_r$; their current expert actions differ by $KA^{r-1}d$. Kernel inclusion is necessary and sufficient for action recovery. When $r \ge h$, left inversion and propagation give R_r. Applying $KF^k x_b$ produces $x_{b+k} = F^k x_b$ by induction. □

Thus observability removes a representational need for chunking, not a statistical learning difficulty. Noiseless expert histories have rank at most n; narrow histories can have rank one. More context can enlarge the set of unidentified coefficients. Under shared process disturbances, one-step tracking obeys $d_{t+1} = Fd_t + Be_t$, where $e_t = \hat{\pi}(z_t) - Kx_t$. If $\|F^k\| \le M\beta^k$ and deployment error is at most ε, then $\limsup_t \|d_t\| \le M\|B\|\varepsilon/(1-\beta)$. Training MSE alone does not supply this deployment bound.

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2. Chunk stability and a conditional history tradeoff

Proposition 2 (Horton's history-based chunk guarantee). Let $\|z\|_{\mathcal{Z}}$ sum the Euclidean norms of its observation/action blocks. Suppose all admissible trajectory pairs satisfy

$$\|\delta y_{t+k}\| \le C_0 \rho^k \|\delta z_t\|_{\mathcal{Z}} + C_0 \sum_{i<k} \rho^{k-1-i} \|\delta u_{t+i}\|,$$

with $C_0 \ge 1$ and $0 < \rho < 1$. Use the same policy and schedule on both trajectories, $u_{j\ell+k} = \pi_k(z_{j\ell}) + v_{j\ell+k}$, with $\mathrm{Lip}_{\mathcal{Z}}(\pi_k) \le D\rho^k$. For $0 < \alpha < 1$, set

$$\bar{\rho} = \rho^{1-\alpha}, \quad C_r = 2rC_0\rho^{-(r-1)}, \quad \bar{C} = C_0\left(1 + \frac{D}{e\alpha\rho\log(1/\rho)}\right).$$

If $\ell \ge \max(1, r-1)$ and $C_r(1 + D\ell/\rho)\rho^{\alpha\ell} \le 1$, then

$$\|\delta y_t\| \le \bar{C}\bar{\rho}^t \|\delta z_0\|_{\mathcal{Z}} + \bar{C}C_r \sum_{i<t} \bar{\rho}^{t-1-i} \|\delta v_i\|.$$

Proof sketch. With $Z_j = \|\delta z_{j\ell}\|_{\mathcal{Z}}$ and $V_i = \|\delta v_i\|$, substitute $\|\delta u_{j\ell+k}\| \le D\rho^k Z_j + V_{j\ell+k}$ into the assumed bound. Summing the blocks of the next history gives

$$Z_{j+1} \le C_r(1 + D\ell/\rho)\rho^\ell Z_j + C_r \sum_{i<\ell} \rho^{\ell-1-i} V_{j\ell+i}.$$

The first coefficient is at most $\bar{\rho}^\ell$. Iterate and use $k\rho^{\alpha k} \le [e\alpha\log(1/\rho)]^{-1}$ within chunks. □

For observable Schur-stable linear plants with $r \ge h$, reconstruction yields the assumed output bound, with suitable constants. This makes Horton's argument complete, but it *does not prove that context shortens execution*: C_r grows with r, and $\ell \ge r-1$ cannot certify $r = 16, \ell = 1$.

Proposition 3 (conditional certificate for general learners). Choose a norm with $\|F\| \le a < 1$. Suppose an arbitrary deterministic learner satisfies, on every admissible deployment history,

$$\left\|\hat{\pi}_{r,j}(z_t) - KF^j x_t\right\| \le \gamma_r q^j \|x_t\|, \qquad 0 < q < 1.$$

A sufficient condition for exponential stability is

$$G(r,\ell) := a^\ell + \gamma_r \sum_{j<\ell} \left\|A^{\ell-1-j}B\right\| q^j < 1.$$

Proof. Writing the prediction residual as e_j, unrolling gives $x_{t+\ell} = F^\ell x_t + \sum_{j<\ell} A^{\ell-1-j}Be_j$; hence $\|x_{t+\ell}\| \le G(r,\ell)\|x_t\|$. Finite within-chunk gains extend decay to all phases. □

If γ_r decreases, certified feasible execution sets expand; their shortest tested length cannot increase. In particular $\gamma_r \|B\| < 1 - a$ certifies $\ell = 1$. If A is also stable and the envelope extends to arbitrary H, $G(r,\ell) \to 0$ as $\ell \to \infty$. The relative-error assumption is strong and is *not established* for our raw-history ridge learner. Neither observability nor additional features guarantees decreasing deployment error. The certificate concerns stability, not monotone tracking loss.

3. A specific mechanism for the narrow-data ridge learner

For the challenging pair of Simchowitz et al. (2025, Def. 4.1) at $\mu = 1/8$, with $B = I$, our experiment fixes

$$A_1 = \begin{bmatrix} 1+\mu & c \\ -c & 1-2\mu \end{bmatrix}, \quad K_1 = \begin{bmatrix} -(1+\mu) & -c \\ c & 0 \end{bmatrix}, \quad c = 3\mu/2.$$

Both A_1 and $A_1 + K_1$ are stable. On the demonstration support $x_1 = 0$, $x_{2,t+1} = qx_{2,t}$ and $u_{1,t} = -cx_{2,t}$, with $q = 3/4$ and $c = 3/16$. Every feature is $x_{2,t}v_r$: its observation coefficients are q^{-k} in the x_2 slots, $0 \le k < r$, and past-action coefficients are $-cq^{-k}$ in the u_1 slots, $1 \le k < r$; all other slots are zero.

Let $S = \sum_i x_{2,t_i}^2 > 0$, $\delta = \lambda/S$, and

$$D_r = \|v_r\|_2^2 = 1 + (1+c^2)\sum_{k=1}^{r-1} q^{-2k}, \qquad b_H = \mathrm{col}((-cq^j, 0))_{j=0}^{H-1}.$$

Ridge regression minimizes $\left\|XW^\top - Y\right\|_F^2 + \lambda \|W\|_F^2$. Since $X = sv_r^\top$ and $Y = sb_H^\top$, its normal equations give exactly

$$\widehat{W} = \frac{b_H v_r^\top}{D_r + \delta}.$$

Thus one-step execution uses

$$u_{1,t} = -\frac{c}{D_r+\delta}\sum_{k=0}^{r-1} q^{-k}x_{2,t-k} + \frac{c^2}{D_r+\delta}\sum_{k=1}^{r-1} q^{-k}u_{1,t-k}, \qquad u_{2,t} = 0.$$

History changes the ridge-selected feedback outside the demonstrations; it does not increase their rank or identify the missing expert correction to x_1.

Proposition 4 (history can stabilize one-step ridge control). Define

$$\alpha_r = \frac{c\sum_{k=0}^{r-1} q^{-k}}{D_r+\delta}, \qquad \beta_r = \frac{c^2\sum_{k=1}^{r-1} q^{-k}}{D_r+\delta}.$$

The one-step controller is asymptotically stable on A_1 if

$$48\alpha_r + \beta_r < 1.$$

Proof. Write $A_1 = aI + N$, $a = 15/16$, where $N^2 = 0$ and $N_{21} = -c$. The absolute impulse-response gain from u_1 to x_2 is

$$\sum_{k\ge 0} |\mathbf{e}_2^\top A_1^k \mathbf{e}_1| = c\sum_{k\ge 1} ka^{k-1} = \frac{c}{(1-a)^2} = 48.$$

The filter coefficient sums are α_r and β_r. The controller reads only x_2 and past u_1 ($u_2 = 0$), so all feedback passes through u_1 with strictly causal ℓ_∞ gain at most $48\alpha_r + \beta_r$; below one, the Neumann series gives a stable inverse, and every component (Schur A_1, shift buffers) is stable. □

Because $D_r = \Theta(q^{-2r})$, both gains are $O(q^r)$: sufficiently long history stabilizes one-step execution. At $\delta = 0$, the certificate is 4.9066, 1.8536, and 0.2021 for $r = 4, 8, 16$. Hence $r = 16$ has an analytic certificate; failing the sufficient test at $r = 8$ proves nothing by itself. Predicted chunks satisfy $\hat{u}_{t+j|t} = q^j \hat{u}_{t|t}$; longer commitment can also stabilize this decaying predictor on the stable plant. Neither mechanism guarantees monotone tracking loss or identification of the expert.

4. Evidence, interpretation, and limits

The earlier noisy midterm experiment ($n = 10, h = 4$) found that context three required two-step commitment, with tail MSE 2.80×10^{-4} ($\ell = 4, 8$ also diverged), while context four enabled one-step execution at 7.64×10^{-6}. These retained results were not rerun here.

The current full-observation sweep ($h = 1$) evaluates 75 (r, H, ℓ) configurations over five training seeds (375 runs), using 32 narrow noiseless demonstrations per seed, $\lambda = 10^{-6}$, and matched target times across r, H. All controllers share 20 held-out initial-state pairs: x_2 uniform on $[-1, 1]$, with $x_1 = 0$ or a signed departure of magnitude 0.0005–0.0015. After a common 15-step expert history, the departure is injected; there is no continuing noise. Test MSE averages full-state tracking error over 200 post-action states and the 20 perturbed rollouts. Separate 1000-step sample losses diagnose long-term growth.

At $H = 16$, median test MSE falls from 2.260×10^{-6} at $(r, \ell) = (8, 8)$ to 6.487×10^{-7} at $(16, 1)$; worst per-step spectral rates are 0.99544 and 0.97110.

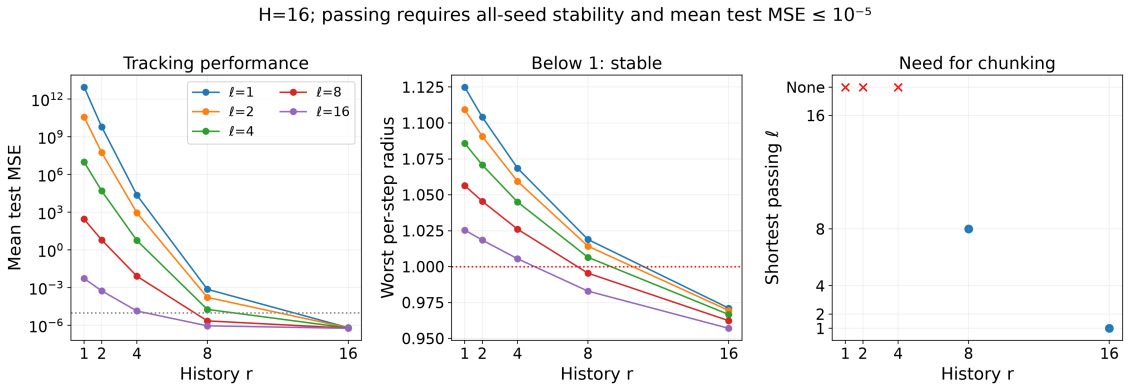


Figure 1: At $H = 16$, history improves test loss and spectral stability. Shortest passing execution is 8 at $r = 8$ and 1 at $r = 16$; None means no tested setting passes. Passing requires stability and mean test MSE $\leq 10^{-5}$ for every seed.

Conclusion. History sixteen permits stable one-step execution, outperforming zero control (MSE 0.08217), though the model-informed memory witness is better (1.736×10^{-8}). At $r = 16$, $\ell = 16$ is still slightly better (5.776×10^{-7}).Since $h = 1$, this reflects feedback selection, not hidden-state recovery. Earlier negative results used continuing noise and $r \leq 8$. The threshold is illustrative; five seeds are descriptive. Generalization requires deployment-error guarantees, not training fit or observability alone.

References

M. Simchowitz, D. Pfrommer, and A. Jadbabaie. The pitfalls of imitation learning when actions are continuous. *arXiv:2503.09722*, 2025.

T. T. Zhang, D. Pfrommer, C. Pan, N. Matni, and M. Simchowitz. Action chunking and exploratory data collection yield exponential improvements in behavior cloning for continuous control. *arXiv:2507.09061*, 2025.